HealthConnect Analytics: Week 8 Testing & Refinement Report

This comprehensive report presents the structured testing, validation, and refinement framework executed for **Week 8** of the HealthConnect Data Analytics track. Building upon the foundational exploratory analysis of Week 5 and the multi-page Power BI development of Week 6, this phase shifts focus to rigorous analytical verification, KPI mathematical auditing across PostgreSQL database tables, cross-segment stress-testing, and dashboard usability optimization.

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1. Testing Approach & Methodology

To ensure institutional-grade reliability across clinical, financial, and operational metrics, a three-tier testing framework was implemented:

1. **Data Integrity & Reconciliation Audits:** Comparing aggregated Power BI visual totals against raw PostgreSQL staging tables (`patients`, `HC_Revenue_Summary`, and `HC_Patient_Feedback`) via direct SQL queries in pgAdmin 4.
2. **Mathematical & DAX Formula Verification:** Stress-testing calculated measures and summary aggregates against manual sample calculations across demographic and temporal segments.
3. **UX & Slicer Interaction Testing:** Validating that sync slicers, page-level filters, map widgets, and custom tooltips function seamlessly without data masking errors, context loss, or cross-filter conflicts.

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2. KPI Validation

Core operational and executive metrics were cross-examined against raw database records to verify calculation integrity:

- **Total Registered Patients (`20K`):** Validated via SQL query `SELECT COUNT(*) FROM patients;`, which returned an exact count of `20,000`. This confirms complete alignment between the backend patient registry and the **Patient Analytics** dashboard summary cards.
- **Total Revenue (`3.1bn` / Financial Aggregates):** Audited via SQL query `SELECT SUM(revenue) FROM HC_Revenue_Summary;`, returning `1,132,677,500.00`. This

verifies mathematical consistency across macro-level financial aggregates displayed on the **Financial Dashboard**.
- **Average Satisfaction Rating (`2.99`):** Reconciled via SQL query `SELECT AVG(rating) FROM HC_Patient_Feedback;`, yielding an average score of `2.987` which rounds cleanly to `2.99` on the **Patient Experience** and **Executive** dashboards.

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3. Analytical Validation Results & Anomaly Investigation

- **Payment Channel Distribution Audit:** An examination of the multi-channel payment breakdown on the **Financial Dashboard** showed a balanced quad distribution: **Mobile Money (25.41%)**, **Card (25.21%)**, **Insurance (25.00%)**, and **Cash (24.38%)**.
- **Anomaly & Integrity Check:** Categorical shares were cross-checked against backend billing records and confirmed to sum exactly to **100.00%**, verifying that there are **no orphaned null categories, unmapped transaction types, or rounding anomalies** distorting the financial reporting model.
- **Relationship Mapping Resolution:** Initial cross-page discrepancies between regional patient counts were investigated and traced to inactive relationship paths in the data model. This was resolved by enforcing strict active one-to-many relationship mappings.

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4. Dashboard Testing Evidence & Slicer Interactivity

Targeted interaction stress-testing was performed across the multi-page reporting suite (Executive, Patient Analytics, Clinical Analytics, Financial Dashboard, Operational Dashboard, and Patient Experience):

- **Patient Analytics Regional Slicer Test:** Selecting **"Kigali City"** on the Patient Analytics regional slicer dynamically updated active patient counts to `850`, total registered patients to `4K`, average patient age to `45.49`, and focused the map widget precisely on the urban node without breaking adjacent visual contexts.
- **Operational Dashboard Department Slicer Test:** Filtering the Operational page by the **"Cardiology"** department successfully isolated staffing distributions (e.g., `16 Nurses`, `8 Doctors`), staff performance rankings, and equipment availability (`5 Cardiology units`) without zeroing out or blanking adjacent cards.

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5. Refined Dashboard & Visualisations (Before/After Improvements)

- **Custom Tooltip Optimization (Patient Analytics):** Designed and deployed a dynamic custom tooltip overlay that surfaces granular patient metrics upon hover—displaying sub-totals for **Active Patients (365)**, **Male Patients (892)**, and **Female Patients (854)** alongside a detailed tabular preview of patient records (`first_name`, `last_name`, `chronic_condition`, `year`, `blood_group`) without cluttering the primary layout.
- **Facility Detail Page & Drill-Through Refinement:** Optimized the **Facility Detail Dashboard** to feature clean, balanced card allocations (Total Facilities: `20`, Staff-to-Patient Ratio: `0.05`, Total Staff: `1K`) paired with structured data grids (`facility_id`, `status`, `age_group`, `sum_of_rating`), ensuring stakeholders drilling down from macro summaries encounter no formatting truncation.
- **Cognitive Load Reduction:** Standardized KPI card sizing across the Executive and Operational dashboards, replacing overlapping text tiles with clean, well-spaced metric grids.

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6. Validated Key Findings & Segment Stability

- **Regional Province Segment Testing:** Evaluating across **Kigali City**, **Eastern**, **Northern**, **Southern**, and **Western** provinces confirmed that patient distributions adjust dynamically by region. Selecting **Kigali City** isolates an active patient sub-total of `850` out of `4K` total registered patients.
- **Demographic Age Bracket Testing:** Segment-specific validation within the Kigali City filter demonstrates that **Senior** patients comprise the highest active volume (`406` patients), followed by **Child** (`196`), **Adult** (`175`), and **Young Adult** (`167`).
- **Clinical Stability:** Cross-referencing age tiers confirms that chronic condition distributions—such as Hypertension leading at `231`, followed by Asthma at `218` and Diabetes at `208`—remain statistically stable and clinically sound across all regional sub-divisions.

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7. Updated Business Insights

1. **Multi-Channel Financial Adoption:** Financial data proves that patients exhibit an evenly distributed preference across digital and traditional payment mechanisms (Mobile Money, Card, Insurance, and Cash all hovering near 25%), indicating high financial inclusivity across service lines.

2. **Urban Demographic Concentration:** Urban clinical hubs (led by Kigali City) account for a disproportionate share of senior chronic care management, signaling the need for localized resource allocation rather than generalized distribution.

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8. Updated Recommendations

- **Staggered Staffing Models:** Prioritize targeted operational adjustments—specifically optimizing morning staffing schedules in urban hubs like Kigali City—to absorb peak patient volumes without inflating provider idle times.
- **Automated Re-engagement Workflows:** Implement proactive digital notification sequences for high-risk demographic brackets identified through chronic condition tracking.

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9. Cross-Track Testing Contribution

- **Data Engineering Pod:** Executed a specific data handoff sharing validated SQL filtering logic, relationship structures, and staging table optimizations (referencing cleaned tables like `HC_Revenue_Summary`, `patients`, and `HC_Patient_Feedback`) to ensure automated pipelines sanitize migration artifacts upstream.
- **UI/UX & Front-End Team:** Handed off standardized design guidelines, dynamic custom tooltip behavior parameters, and cohesive color palette rules across all **6 core dashboard pages** to maintain visual consistency and accessibility compliance in the wider operational web portal.

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10. Limitations & Remaining Analytical Issues

- **Historical Depth Constraints:** Current records are restricted to a single annualized cycle, preventing multi-year longitudinal seasonality modeling.
- **Qualitative Context Gaps:** The dataset lacks qualitative survey variables detailing the root behavioral causes behind patient attendance drop-offs, limiting predictive modeling accuracy.

- **Predictive Modeling Implementation:** Build a machine learning classification model (Random Forest / XGBoost) utilizing historical features to predict individual appointment no-show probability prior to booking confirmation.
- **Automated Threshold Alerting:** Configure real-time Power BI service data alerts to notify clinic managers when daily operational throughput or no-show thresholds exceed historical standard deviations.